

CSCE 679 DATA VISUALIZATION · MODULE 08 · LECTURE NOTES

Maps

Where position is taken, and every projection lies.

MODULE OUTCOMES

- Explain projection distortion and the choice it forces.
- Explain why area encodings on maps mislead.
- Explain normalisation for spatial data.
- Explain the alternatives to a choropleth.
- Choose a spatial representation for a task.

1 Projection

NO PROJECTION PRESERVES AREA, ANGLE, AND DISTANCE TOGETHER — THIS IS A THEOREM

A sphere cannot be mapped to a plane without distortion — it has intrinsic Gaussian curvature and the plane does not — which is Gauss's theorema egregium and is a mathematical fact rather than an engineering limitation that better algorithms will overcome.

So every projection chooses what to preserve: a conformal projection keeps angles locally and distorts areas; an equal-area projection keeps areas and distorts shapes; and compromise projections do neither exactly but spread the error more evenly.

Which means the choice is determined by the task (Module 01 §2 again): navigation wants angles preserved, and any figure that encodes a quantity on area wants equal area — and using a conformal projection for the latter is a measurable error rather than a stylistic preference.

And the familiar world projection is conformal — Mercator, chosen for navigation — so it inflates the high latitudes severely, by a large factor near the poles, which matters whenever a reader compares the visual sizes of countries and is why so many people's sense of relative national areas is wrong.

- **Equal-area for anything quantitative** — because a reader comparing two regions' shaded areas is making an area judgement (Module 02 §1's rank five), and the projection must not already have distorted the areas before the encoding did.
- **Conformal for navigation and for local detail**, where angles and local shapes matter and the extent is small enough that the area distortion is negligible — which is most city and regional mapping.
- **A compromise projection for general reference**, which is what most atlases use and is a reasonable default for a non-quantitative map where no particular property is being read.

- **And a local projection for local data — a country-scale map should use that country's standard national projection**, not a world projection cropped, which distorts unnecessarily.
- **With the projection named in the caption, which almost never happens and should — and is the only way a reader can know what the areas in the figure mean** (§4's closing argument).

2 The choropleth problem

A CHOROPLETH ENCODES A VALUE ON COLOUR AND A REGION'S IMPORTANCE ON ITS AREA

The value is encoded on colour, which is rank six or seven (Module 02 §1) — so **the quantity is read imprecisely, which is tolerable and is the lesser of the two problems**, since a choropleth is usually read for pattern rather than for value.

And the region's visual prominence is determined by its geographic area, which **has nothing whatever to do with its importance, its population, or its contribution to the total** — so **a large sparsely populated region dominates the figure and a dense city is almost invisible**.

Which means a choropleth of a per-capita quantity systematically overweights rural areas in the reader's impression, and **this is a structural property of the representation rather than a styling problem** that better colours would fix.

So the two defects are: an imprecise quantitative channel, and an irrelevant weighting by geographic area — and **the second is the one that changes conclusions**, because it determines what the reader perceives as the dominant pattern.

CARTOGRAM

distort the regions so that area encodes the value. Fixes the weighting; destroys recognisability, and the distortion is itself hard to read quantitatively.

DOT DENSITY

one dot per n units, placed within each region. Area then correctly reflects the total, and visual density reflects the rate. Good, and it needs care about placement so the dots are not read as locations.

PROPORTIONAL SYMBOLS

a circle per region, with area encoding the value. Still rank five (Module 02), and at least the geography is undistorted and recognisable.

SMALL MULTIPLES OF NON-MAPS

a sorted bar chart of the regions, which puts the value on position -- rank one.

AND THAT LAST ONE IS USUALLY BEST, unless the spatial pattern IS the finding. Which is the question to ask: does the reader need the geography?

Does the reader need the geography? — and if not, a sorted bar chart puts the value on position, which is rank one and is read an order of magnitude more accurately than any colour scale. **The does-the-reader-need-geography question is this module's practical core**, and it is §4's first question because it eliminates the majority of maps before any other decision has to be made.

3 Normalisation

A MAP OF RAW COUNTS IS A MAP OF POPULATION, ALMOST WITHOUT EXCEPTION

Any count of people-related events — disease cases, crimes, retail sales, votes, library loans — is roughly proportional to the number of people present, so the map shows where the people are and essentially nothing else.

Which looks very much like a finding, because the pattern is strong, spatially coherent, and reproducible — and it is a finding about population, which you already knew.

So normalise, by population, by area, by exposure, by the number at risk, or by whatever the relevant denominator is — which is Module 03 §3's derivation argument arriving where it matters most and where it is most often skipped.

And then the small-denominator problem appears: a rate computed over a small population is extremely noisy, so the extreme values are all in the smallest regions — which is Module 09's subject and requires showing the uncertainty rather than only the estimate.

- The small-denominator problem — the highest and the lowest rates are both found in the smallest regions, by construction rather than by coincidence, and a naive map highlights noise as though it were signal (Module 09 §4).
- The modifiable areal unit problem — the same underlying data aggregated to different boundaries gives visibly different patterns, and the boundaries were chosen for administrative reasons unrelated to your question.
- Which means the choice of spatial unit is an analytical decision, and aggregating to a different level is a legitimate and reportable choice rather than a manipulation — provided it is stated.
- And the binning of the colour scale — quantiles, equal intervals, and natural breaks give visibly different maps from identical data, because they place the class boundaries differently and the boundaries are where the visual contrast is.
- So state the denominator, the units, and the binning, all three — plus the projection from §1. The binning choice is as consequential as the normalisation, and it is even less often reported.

4 Choosing

- Is the spatial pattern the finding? If not, do not use a map — use a chart that puts the value on position (§2's last alternative).
- Is the quantity a rate or a count? If it is a count, either normalise it or use dot density, where area correctly reflects the total rather than misrepresenting it.
- Does the reader need to recognise the places? If yes, a cartogram is out — and if no, the cartogram's main cost disappears.

- **Is the denominator small anywhere? If so, the uncertainty has to be shown** (Module 09) — or the small regions will dominate the reader's impression with noise.
- **And which projection does the task require? (§1) — equal-area for anything quantitative**, and named in the caption. **'Is the spatial pattern the finding' is the first question and the one that eliminates most maps — a great many maps exist because the data happened to have a location column**, which is not a reason.

AND A MAP IS UNUSUALLY PERSUASIVE, WHICH RAISES THE STANDARD

A map is read as authoritative and concrete in a way that a bar chart is not — it looks like a photograph of the world rather than an encoding of a table, and readers extend to it the credence they give to a geographic fact.

Which means its errors are believed, and a badly normalised choropleth produces confident wrong conclusions about places and about the people who live in them — which is a consequence beyond the figure (CSCE 638 Module 12's framing, arriving visually).

So the normalisation, the spatial units, the colour binning, and the projection all have to be stated — four things, and a typical published map states none of them.

Which is this module's version of the program's rule — and the four statements are the difference between a figure a reader can assess and one they can only believe, which is Module 13's distinction exactly.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Munzner, chapter 8 (slides free)	§2's alternatives — the spatial representation design space, with the trades.
Brewer — Designing Better Maps	§1 and §3 — projections, classification, and colour for maps, from a cartographer. Library copy.
Openshaw — The Modifiable Areal Unit Problem	§3's second trap — the original statement, and it is a more serious problem than its obscurity suggests. Library copy.
Wilke, the geospatial chapters (free)	§1 and §2 practically , with the projection comparisons drawn out.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Render the same world data in Mercator and in an equal-area projection**, and compare the apparent sizes.
- 2 **Measure the area inflation** at 60 degrees latitude.
- 3 **Find a published choropleth** and identify the largest region and its population.
- 4 **Map raw counts of something population-driven**, then map the rate, and compare.
- 5 **Build a dot density map** of the same data.
- 6 **Build a sorted bar chart** of the same data and ask five people to rank three regions from each.
- 7 **Find the extreme-rate regions** and check their denominators.
- 8 **Re-bin one choropleth three ways** and compare the maps.
- 9 **Aggregate the same data to two different unit levels** and compare the patterns.
- 10 **Write the four required statements** for a map you made.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why can no projection preserve everything, and what kind of claim is that?
- 2 Which projection for quantitative data, and why?
- 3 Give the four projection choices by purpose.
- 4 Name the two defects of a choropleth, and which changes conclusions.
- 5 Give four alternatives and say which is usually best.
- 6 Why is a map of counts a map of population?
- 7 What is the small-denominator problem?
- 8 What is the modifiable areal unit problem, and what follows?
- 9 Give the five decision questions in order.
- 10 Why does a map's persuasiveness raise the standard, and what four things must be stated?